

requires-clause syntax ambiguities: wording to apply custom grammar

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What's New

The previous revision of the paper, P0730R0, was discussed in EWG on Tuesday of the 2017 Toronto meeting of WG 21. Direction was given to pursue the custom grammar solution with wording to present to Core. This paper, P0730R1, is the wording paper.

Background and Context for This Paper

Prior to the discussion of P0730R0 in EWG, direction was given in response to P0716R0, which removes function concepts. The wording presented herein is contingent upon adopting wording that implements the removal of function concepts.

Wording

The editing instructions within this document use, as its base wording, the result of applying the TS working draft (N4674) to an appropriate base document.

In Clause 17 [temp] paragraph 1, modify:

template-declaration:

template < *template-parameter-list* > *requires-clause*_{opt} *declaration*
template-introduction declaration

~~*requires-clause:*~~

~~*requires constraint-expression*~~

template-parameter-list:

template-parameter
template-parameter-list , *template-parameter*

requires-clause:

requires constraint-logical-or-expression

constraint-logical-and-expression:

primary-expression
constraint-logical-and-expression && *primary-expression*

constraint-logical-or-expression:

constraint-logical-and-expression
constraint-logical-or-expression | constraint-logical-and-expression

At the beginning of subclause 8.1.7.4 [expr.prim.req.nested], modify:

nested-requirement:
~~requires-clause~~ requires constraint-expression ;

In subclause 17.10.2 [temp.constr.decl] paragraph 2, modify bullet 2.3.5:

- the *constraint-expression* of introduced by a trailing *requires-clause* of a function declaration.

In subclause 17.10.1.2 [temp.constr.pred] paragraph 1, modify the note:

[...] [*Note:* A predicate constraint is introduced by ~~the constraint-expression of a~~ *requires-clause* (17.10.2), or as the associated constraint of a *constrained-parameter* (17.1) or *template-introduction* (17.2). —end note] [...]

In Clause 17 [temp] paragraph 7, modify:

A *template-declaration* is written in terms of its template parameters. These parameters are declared explicitly in a *template-parameter-list* (17.1), or they are introduced by a *template-introduction* (17.2). The optional *requires-clause* following a *template-parameter-list* allows the specification of constraints (17.10.2) on template arguments (17.4). The *requires-clause* introduces the *constraint-expression* that results from interpreting the *constraint-logical-or-expression* as a *constraint-expression*.

Acknowledgements

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